

CLASS 10 · ICSE · SAMPLE PAPER SOLUTIONS

# MathSetu Sample Paper 1: Mathematics

Step-by-step solutions with the marks for each question

The question numbers match the question paper.

## 1 Section A, Question 1: choose the correct answer (15 marks)

1. An article is marked at ₹800 and sold within the same state. If the rate of GST is 18%, the CGST charged is: (a) ₹36 (b) ₹72 (c) ₹80 (d) ₹144 **1 mark**

SOLUTION

For a sale within a state, CGST is half the GST rate: 9% of ₹800 = ₹72.

**Answer: (b) ₹72**

2. ₹500 is deposited every month for 12 months in a recurring deposit account. The equivalent principal for one month, used to calculate the interest, is: (a) ₹6000 (b) ₹33,000 (c) ₹39,000 (d) ₹78,000 **1 mark**

SOLUTION

Equivalent principal for one month =  $P \times \frac{n(n+1)}{2} = 500 \times \frac{12 \times 13}{2} = ₹39,000$ .

**Answer: (c) ₹39,000**

3. A ₹100 share is bought at ₹125 and pays a dividend of 10%. The rate of return on the investment is: (a) 8% (b) 10% (c) 12.5% (d) 25% **1 mark**

SOLUTION

Dividend on one share = ₹10 on an investment of ₹125, so the return is  $\frac{10}{125} \times 100 = 8\%$ .

**Answer: (a) 8%**

4. The solution set of  $-3 < 2x - 1 \leq 5$ , where  $x$  is a whole number, is: (a)  $\{1, 2, 3\}$  (b)  $\{0, 1, 2\}$  (c)  $\{0, 1, 2, 3\}$  (d)  $\{-1, 0, 1, 2, 3\}$  **1 mark**

SOLUTION

Add 1 throughout:  $-2 < 2x \leq 6$ , so  $-1 < x \leq 3$ . The whole numbers in this range are 0, 1, 2, 3.

**Answer: (c)  $\{0, 1, 2, 3\}$**

5. The roots of  $x^2 - 6x + 9 = 0$  are: (a) real and unequal (b) real and equal (c) not real (d) equal in size and opposite in sign **1 mark**

SOLUTION

$D = (-6)^2 - 4(1)(9) = 0$ , so the roots are real and equal (both are 3).

**Answer: (b) real and equal**

6. The mean proportional between 4 and 25 is: (a) 10 (b) 14.5 (c) 21 (d) 100 **1 mark**

SOLUTION

Mean proportional =  $\sqrt{4 \times 25} = 10$ .

**Answer: (a) 10**

7. The remainder when  $x^3 - 2x^2 + 3x - 4$  is divided by  $x - 2$  is: (a) -4 (b) 0 (c) 2 (d) 6 **1 mark**

SOLUTION

By the Remainder Theorem, the remainder is  $f(2) = 8 - 8 + 6 - 4 = 2$ .

**Answer: (c) 2**

8.  $A$  is a matrix of order  $2 \times 3$  and  $B$  is a matrix of order  $3 \times 1$ . The order of  $AB$  is: (a)  $3 \times 3$  (b)  $2 \times 1$  (c)  $1 \times 2$  (d)  $AB$  is not possible **1 mark**

SOLUTION

The columns of  $A$  (3) equal the rows of  $B$  (3), so  $AB$  exists, with the rows of  $A$  and the columns of  $B$ :  $2 \times 1$ .

**Answer: (b)  $2 \times 1$**

9. The  $n$ th term of an A.P. is  $5n - 2$ . Its common difference is: (a) -2 (b) 2 (c) 3 (d) 5 **1 mark**

SOLUTION

$T_1 = 3$  and  $T_2 = 8$ , so  $d = 8 - 3 = 5$ .

**Answer: (d) 5**

10. The 5th term of the G.P. 3, 6, 12, ... is: (a) 24 (b) 36 (c) 48 (d) 96 **1 mark**

SOLUTION

$a = 3, r = 2$ :  $T_5 = ar^4 = 3 \times 16 = 48$ .

**Answer: (c) 48**

- 11.** The image of the point  $(3, -4)$  after reflection in the  $y$ -axis is: (a)  $(3, 4)$  (b)  $(-3, 4)$  (c)  $(-3, -4)$  (d)  $(-4, 3)$  **1 mark**

SOLUTION

Reflection in the  $y$ -axis changes the sign of the  $x$ -coordinate only:  $(3, -4) \rightarrow (-3, -4)$ .

**Answer: (c)**  $(-3, -4)$

- 12.** The slope of the line  $3x - 4y + 7 = 0$  is: (a)  $\frac{3}{4}$  (b)  $-\frac{3}{4}$  (c)  $\frac{4}{3}$  (d)  $\frac{7}{4}$  **1 mark**

SOLUTION

Write it as  $y = \frac{3}{4}x + \frac{7}{4}$ . The slope is  $\frac{3}{4}$ .

**Answer: (a)**  $\frac{3}{4}$

- 13.** An arc of a circle subtends an angle of  $110^\circ$  at the centre. The angle it subtends at a point on the remaining part of the circle is: (a)  $55^\circ$  (b)  $70^\circ$  (c)  $110^\circ$  (d)  $220^\circ$  **1 mark**

SOLUTION

The angle at the centre is twice the angle at the circumference:  $\frac{110^\circ}{2} = 55^\circ$ .

**Answer: (a)**  $55^\circ$

- 14.** The volume of a cone of radius 3 cm and height 7 cm is: (a)  $22 \text{ cm}^3$  (b)  $66 \text{ cm}^3$  (c)  $132 \text{ cm}^3$  (d)  $198 \text{ cm}^3$  **1 mark**

SOLUTION

Volume =  $\frac{1}{3}\pi r^2 h = \frac{1}{3} \times \frac{22}{7} \times 9 \times 7 = 66 \text{ cm}^3$ .

**Answer: (b)**  $66 \text{ cm}^3$

- 15.** A letter is chosen at random from the letters of the word TRIANGLE. The probability that it is a vowel is: (a)  $\frac{1}{4}$  (b)  $\frac{3}{8}$  (c)  $\frac{1}{2}$  (d)  $\frac{5}{8}$  **1 mark**

SOLUTION

The word has 8 letters, of which 3 (I, A, E) are vowels. Probability =  $\frac{3}{8}$ .

**Answer: (b)**  $\frac{3}{8}$

## 2 Section A, Question 2 (12 marks)

16. Meera deposits ₹1200 every month for 2 years in a recurring deposit account that pays simple interest at 8% per annum. Find the interest she earns and the amount she gets on maturity.

4 marks

SOLUTION

$$P = ₹1200, n = 24 \text{ months}, r = 8\%.$$

$$\text{Interest} = P \times \frac{n(n+1)}{2 \times 12} \times \frac{r}{100} = 1200 \times \frac{24 \times 25}{24} \times \frac{8}{100} = ₹2400.$$

$$\text{Total deposited} = 1200 \times 24 = ₹28,800, \text{ so the maturity amount} = 28,800 + 2400 = ₹31,200.$$

**Answer: interest ₹2400, maturity amount ₹31,200**

17. If  $A = \begin{bmatrix} 2 & 1 \\ -1 & 3 \end{bmatrix}$  and  $B = \begin{bmatrix} 1 & 0 \\ 2 & -1 \end{bmatrix}$ , find  $A^2 - 2B$ . 4 marks

SOLUTION

$$A^2 = \begin{bmatrix} 2 & 1 \\ -1 & 3 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ -1 & 3 \end{bmatrix} = \begin{bmatrix} 4-1 & 2+3 \\ -2-3 & -1+9 \end{bmatrix} = \begin{bmatrix} 3 & 5 \\ -5 & 8 \end{bmatrix}$$

$$2B = \begin{bmatrix} 2 & 0 \\ 4 & -2 \end{bmatrix}$$

$$A^2 - 2B = \begin{bmatrix} 3-2 & 5-0 \\ -5-4 & 8+2 \end{bmatrix}$$

$$\text{Answer: } \begin{bmatrix} 1 & 5 \\ -9 & 10 \end{bmatrix}$$

18. Prove that  $\frac{\cos A}{1 - \sin A} + \frac{\cos A}{1 + \sin A} = 2 \sec A$ . 4 marks

SOLUTION

$$\text{LHS} = \frac{\cos A (1 + \sin A) + \cos A (1 - \sin A)}{(1 - \sin A)(1 + \sin A)}$$

$$= \frac{2 \cos A}{1 - \sin^2 A} = \frac{2 \cos A}{\cos^2 A}$$

$$= \frac{2}{\cos A} = 2 \sec A = \text{RHS. Proved.}$$

## 3 Section A, Question 3 (13 marks)

19. A solid metal sphere of radius 6 cm is melted and recast into a solid cylinder of radius 4 cm. Find the height of the cylinder and its curved surface area, correct to one decimal place. 4 marks

SOLUTION

$$\text{Volume of the sphere} = \frac{4}{3} \pi \times 6^3 = 288\pi \text{ cm}^3.$$

$$\text{Volume of the cylinder} = \pi \times 4^2 \times h = 16\pi h. \text{ So } 16\pi h = 288\pi \text{ and } h = 18 \text{ cm.}$$

$$\text{Curved surface area} = 2\pi rh = 2 \times \frac{22}{7} \times 4 \times 18 = \frac{3168}{7} \approx 452.6 \text{ cm}^2.$$

**Answer: height 18 cm, curved surface area  $\approx 452.6 \text{ cm}^2$**

- 20.**  $A(2, 3)$  and  $B(6, -5)$  are two points. Find (i) the slope of  $AB$  (ii) the equation of the perpendicular bisector of  $AB$  (iii) the point where this bisector meets the  $x$ -axis. **4 marks**

**SOLUTION**

(i) Slope of  $AB = \frac{-5 - 3}{6 - 2} = -2$ .

(ii) The mid-point of  $AB$  is  $\left(\frac{2+6}{2}, \frac{3-5}{2}\right) = (4, -1)$ , and the slope of the perpendicular bisector is  $\frac{1}{2}$ .

Its equation is  $y + 1 = \frac{1}{2}(x - 4)$ , that is  $x - 2y - 6 = 0$ .

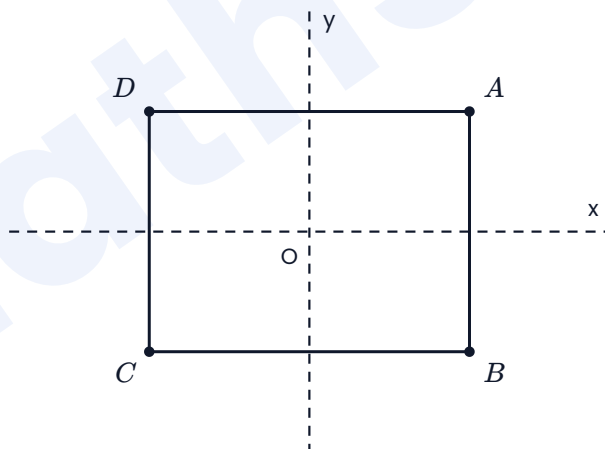
(iii) Put  $y = 0$ :  $x = 6$ . The point is  $(6, 0)$ .

**Answer: (i)  $-2$  (ii)  $x - 2y - 6 = 0$  (iii)  $(6, 0)$**

- 21.** Use graph paper for this question, with  $1 \text{ cm} = 1 \text{ unit}$  on both axes. Plot the point  $A(4, 3)$ . Reflect  $A$  in the  $x$ -axis to get  $B$ , reflect  $A$  in the origin to get  $C$ , and reflect  $A$  in the  $y$ -axis to get  $D$ . (i) Write the coordinates of  $B$ ,  $C$  and  $D$ . (ii) Name the figure  $ABCD$ . (iii) Find its perimeter and its area. **5 marks**

**SOLUTION**

(i)  $B = (4, -3)$ ,  $C = (-4, -3)$  and  $D = (-4, 3)$ .



(ii)  $AB$  and  $CD$  are vertical,  $BC$  and  $DA$  are horizontal, and all the angles are right angles, so  $ABCD$  is a rectangle.

(iii)  $AB = 6$  units and  $BC = 8$  units. Perimeter  $= 2(6 + 8) = 28$  units and area  $= 6 \times 8 = 48$  square units.

**Answer: (i)  $B(4, -3)$ ,  $C(-4, -3)$ ,  $D(-4, 3)$  (ii) a rectangle (iii) perimeter 28 units, area 48 square units**

## 4 Section B, Question 4 (10 marks)

- 22.** A dealer in Delhi sells a washing machine, marked at ₹20,000, to a customer in Delhi at a discount of 10%. The rate of GST is 18%. Find the CGST, the SGST and the total amount the customer pays.

3 marks

**SOLUTION**

Selling price after discount =  $20,000 - 10\% \text{ of } 20,000 = ₹18,000$ .

CGST = 9% of 18,000 = ₹1620 and SGST = 9% of 18,000 = ₹1620.

Amount paid =  $18,000 + 1620 + 1620 = ₹21,240$ .

**Answer: CGST ₹1620, SGST ₹1620, amount paid ₹21,240**

- 23.** Solve  $x^2 - 5x - 3 = 0$ , giving the answers correct to two decimal places. 3 marks

**SOLUTION**

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{5 \pm \sqrt{25 + 12}}{2} = \frac{5 \pm \sqrt{37}}{2}, \text{ and } \sqrt{37} \approx 6.083.$$

$$x = \frac{5 + 6.083}{2} \approx 5.54 \text{ or } x = \frac{5 - 6.083}{2} \approx -0.54.$$

**Answer:  $x = 5.54$  or  $x = -0.54$**

- 24.** Find the mean of the following distribution by the step-deviation method. 4 marks

| Class     | 0 – 20 | 20 – 40 | 40 – 60 | 60 – 80 | 80 – 100 |
|-----------|--------|---------|---------|---------|----------|
| Frequency | 6      | 10      | 16      | 12      | 6        |

**SOLUTION**

Take the assumed mean  $A = 50$  and the class width  $i = 20$ , with  $t = \frac{x - A}{i}$ .

| Class    | $f$ | $x$ | $t$ | $ft$ |
|----------|-----|-----|-----|------|
| 0 – 20   | 6   | 10  | -2  | -12  |
| 20 – 40  | 10  | 30  | -1  | -10  |
| 40 – 60  | 16  | 50  | 0   | 0    |
| 60 – 80  | 12  | 70  | 1   | 12   |
| 80 – 100 | 6   | 90  | 2   | 12   |
| Total    | 50  |     |     | 2    |

$$\text{Mean} = A + \frac{\sum ft}{\sum f} \times i = 50 + \frac{2}{50} \times 20 = 50.8.$$

**Answer: mean = 50.8**

**5 Section B, Question 5 (10 marks)**

- 25.** A man invests ₹9600 in ₹100 shares at a premium of 20%. The company declares a dividend of 15%. Find (i) the number of shares he buys (ii) his annual dividend (iii) the rate of return on his investment. **3 marks**

**SOLUTION**

(i) Market value of one share =  $100 + 20 = ₹120$ . Number of shares =  $\frac{9600}{120} = 80$ .

(ii) Dividend on one share = 15% of ₹100 = ₹15. Annual dividend =  $80 \times 15 = ₹1200$ .

(iii) Rate of return =  $\frac{1200}{9600} \times 100 = 12.5\%$ .

**Answer: (i) 80 shares (ii) ₹1200 (iii) 12.5%**

- 26.** In a cyclic quadrilateral  $ABCD$ ,  $\angle A = (2x + 4)^\circ$ ,  $\angle B = (x + 40)^\circ$  and  $\angle C = (3x + 16)^\circ$ . Find  $x$  and all four angles. **3 marks**

**SOLUTION**

Opposite angles of a cyclic quadrilateral add up to  $180^\circ$ :  $(2x + 4) + (3x + 16) = 180$ , so  $5x = 160$  and  $x = 32$ .

$\angle A = 68^\circ$ ,  $\angle C = 112^\circ$ ,  $\angle B = 72^\circ$  and  $\angle D = 180^\circ - 72^\circ = 108^\circ$ .

**Answer:  $x = 32$ ; the angles are  $68^\circ$ ,  $72^\circ$ ,  $112^\circ$  and  $108^\circ$**

- 27.** Use the Factor Theorem to show that  $x - 2$  is a factor of  $2x^3 + x^2 - 13x + 6$ . Hence factorise the expression completely. **4 marks**

**SOLUTION**

Let  $f(x) = 2x^3 + x^2 - 13x + 6$ . Then  $f(2) = 16 + 4 - 26 + 6 = 0$ , so  $x - 2$  is a factor.

Dividing  $f(x)$  by  $x - 2$  gives the quotient  $2x^2 + 5x - 3$ .

$2x^2 + 5x - 3 = 2x^2 + 6x - x - 3 = 2x(x + 3) - 1(x + 3) = (2x - 1)(x + 3)$ .

**Answer:  $(x - 2)(2x - 1)(x + 3)$**

**6 Section B, Question 6 (10 marks)**

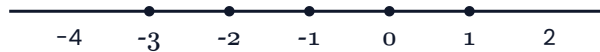
**28.** Solve  $3x - 7 \leq 7x + 5 < 19$ , where  $x$  is an integer, and show the solution set on a number line.

**3 marks**

**SOLUTION**

From  $3x - 7 \leq 7x + 5$ :  $-12 \leq 4x$ , so  $x \geq -3$ . From  $7x + 5 < 19$ :  $7x < 14$ , so  $x < 2$ .

So  $-3 \leq x < 2$ , and the solution set is  $\{-3, -2, -1, 0, 1\}$ . On the number line these five integers are marked with dots.



**Answer:**  $\{-3, -2, -1, 0, 1\}$

**29.** The 4th term of an A.P. is 22 and its 15th term is 66. Find the first term, the common difference and the sum of its first 8 terms. **3 marks**

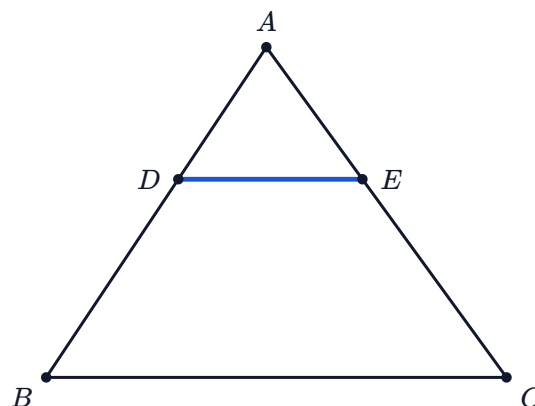
**SOLUTION**

$a + 3d = 22$  and  $a + 14d = 66$ . Subtracting,  $11d = 44$ , so  $d = 4$  and  $a = 10$ .

$$S_8 = \frac{8}{2} [2 \times 10 + 7 \times 4] = 4 \times 48 = 192.$$

**Answer:**  $a = 10, d = 4, S_8 = 192$

- 30.** In  $\triangle ABC$ ,  $D$  lies on  $AB$  and  $E$  lies on  $AC$  so that  $DE \parallel BC$  and  $AD : DB = 2 : 3$ . (i) Prove that  $\triangle ADE \sim \triangle ABC$ . (ii) If  $DE = 4$  cm, find  $BC$ . (iii) Find the ratio of the area of  $\triangle ADE$  to the area of trapezium  $DBCE$ . **4 marks**



**SOLUTION**

(i)  $\angle A$  is common and  $\angle ADE = \angle ABC$  (corresponding angles,  $DE \parallel BC$ ). So  $\triangle ADE \sim \triangle ABC$  (AA).

(ii)  $\frac{AD}{AB} = \frac{2}{2+3} = \frac{2}{5}$ , so  $\frac{DE}{BC} = \frac{2}{5}$  and  $BC = \frac{5}{2} \times 4 = 10$  cm.

(iii)  $\frac{\text{area of } \triangle ADE}{\text{area of } \triangle ABC} = \left(\frac{2}{5}\right)^2 = \frac{4}{25}$ . So the trapezium has  $25 - 4 = 21$  of the 25 parts.

**Answer: (ii)  $BC = 10$  cm (iii)  $4 : 21$**

**7 Section B, Question 7 (10 marks)**

- 31.** What number must be added to each of 2, 6, 8 and 18 so that the four results are in proportion? **3 marks**

**SOLUTION**

Let the number be  $x$ . Then  $\frac{2+x}{6+x} = \frac{8+x}{18+x}$ .

$(2+x)(18+x) = (6+x)(8+x)$  gives  $36 + 20x + x^2 = 48 + 14x + x^2$ , so  $6x = 12$  and  $x = 2$ .

Check: 4, 8, 10, 20 are in proportion, since  $4 \times 20 = 8 \times 10$ .

**Answer: 2**

- 32.** The 3rd term of a G.P. is 12 and its 6th term is 96. Find the first term, the common ratio and the sum of its first 8 terms. **3 marks**

**SOLUTION**

$ar^2 = 12$  and  $ar^5 = 96$ . Dividing,  $r^3 = 8$ , so  $r = 2$  and  $a = 3$ .

$$S_8 = \frac{a(r^8 - 1)}{r - 1} = \frac{3(256 - 1)}{2 - 1} = 765.$$

**Answer:  $a = 3, r = 2, S_8 = 765$**

- 33.** From a point  $A$  on level ground, the angle of elevation of the top of a tower is  $30^\circ$ . On walking 40 m towards the tower to a point  $B$ , the angle of elevation becomes  $60^\circ$ . Find the height of the tower, correct to two decimal places, and the distance of  $B$  from the tower. (Take  $\sqrt{3} = 1.732$ .)

4 marks

SOLUTION

Let the tower be  $h$  m high and let  $B$  be  $x$  m from its foot.

From  $B$ :  $\tan 60^\circ = \frac{h}{x}$ , so  $h = \sqrt{3}x$ . From  $A$ :  $\tan 30^\circ = \frac{h}{x+40}$ , so  $x+40 = \sqrt{3}h$ .

Then  $x+40 = \sqrt{3} \times \sqrt{3}x = 3x$ , so  $x = 20$  m.

$h = 20\sqrt{3} = 20 \times 1.732 = 34.64$  m.

**Answer: height 34.64 m;  $B$  is 20 m from the tower**

## 8 Section B, Question 8 (10 marks)

- 34.** Cards numbered 1 to 30 are put in a box and one card is drawn at random. Find the probability that the number on it is (i) a multiple of 4 (ii) a prime number (iii) a perfect square. **3 marks**

SOLUTION

(i) Multiples of 4: 4, 8, 12, 16, 20, 24, 28, which is 7 cards. Probability =  $\frac{7}{30}$ .

(ii) Primes: 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, which is 10 cards. Probability =  $\frac{10}{30} = \frac{1}{3}$ .

(iii) Perfect squares: 1, 4, 9, 16, 25, which is 5 cards. Probability =  $\frac{5}{30} = \frac{1}{6}$ .

**Answer: (i)  $\frac{7}{30}$  (ii)  $\frac{1}{3}$  (iii)  $\frac{1}{6}$**

- 35.** In what ratio does the  $y$ -axis divide the line segment joining  $A(-4, 1)$  and  $B(6, 6)$ ? Also find the point of division. **3 marks**

SOLUTION

Let the ratio be  $k : 1$ . The  $x$ -coordinate of the point of division is 0:  $\frac{6k-4}{k+1} = 0$ , so  $k = \frac{2}{3}$ . The ratio is 2 : 3.

$y = \frac{2 \times 6 + 3 \times 1}{2 + 3} = \frac{15}{5} = 3$ .

**Answer: ratio 2 : 3, point (0, 3)**

- 36.** A conical tent is 12 m high and the radius of its base is 9 m. Find (i) the cost of the canvas for its curved surface at ₹70 per m<sup>2</sup> (ii) the volume of air inside it, correct to the nearest m<sup>3</sup>. **4 marks**

**SOLUTION**

Slant height  $l = \sqrt{9^2 + 12^2} = 15$  m.

$$(i) \text{ Curved surface area} = \pi r l = \frac{22}{7} \times 9 \times 15 = \frac{2970}{7} \text{ m}^2. \text{ Cost} = \frac{2970}{7} \times 70 = ₹29,700.$$

$$(ii) \text{ Volume} = \frac{1}{3} \pi r^2 h = \frac{1}{3} \times \frac{22}{7} \times 81 \times 12 = \frac{7128}{7} \approx 1018 \text{ m}^3.$$

**Answer: (i) ₹29,700 (ii) about 1018 m<sup>3</sup>**

## 9 Section B, Question 9 (10 marks)

- 37.** The length of a rectangular garden is 4 m more than its breadth, and its area is 96 m<sup>2</sup>. Find its length, its breadth and its perimeter. **3 marks**

**SOLUTION**

Let the breadth be  $x$  m. Then  $x(x + 4) = 96$ , so  $x^2 + 4x - 96 = 0$ .

$(x + 12)(x - 8) = 0$ , so  $x = 8$  (a length cannot be negative).

Breadth = 8 m, length = 12 m, perimeter =  $2(8 + 12) = 40$  m.

**Answer: length 12 m, breadth 8 m, perimeter 40 m**

- 38.** (i) Chords  $AB$  and  $CD$  of a circle meet at a point  $P$  inside the circle. If  $AP = 6$  cm,  $PB = 4$  cm and  $CP = 3$  cm, find  $CD$ . (ii) From a point  $T$  outside another circle, a tangent  $TQ$  and a secant  $TRS$  are drawn, with  $TR = 4$  cm and  $RS = 5$  cm. Find  $TQ$ . **3 marks**

**SOLUTION**

(i)  $AP \times PB = CP \times PD$ , so  $6 \times 4 = 3 \times PD$  and  $PD = 8$  cm. Then  $CD = 3 + 8 = 11$  cm.

(ii)  $TQ^2 = TR \times TS = 4 \times (4 + 5) = 36$ , so  $TQ = 6$  cm.

**Answer: (i)  $CD = 11$  cm (ii)  $TQ = 6$  cm**

- 39.** The marks of 11 students in a test are 12, 18, 25, 9, 30, 21, 15, 27, 33, 6 and 24. Find (i) the median (ii) the lower quartile (iii) the upper quartile (iv) the interquartile range. **4 marks**

**SOLUTION**

In ascending order: 6, 9, 12, 15, 18, 21, 24, 25, 27, 30, 33. Here  $n = 11$ .

(i) Median =  $\left(\frac{n+1}{2}\right)$ th term = 6th term = 21.

(ii) Lower quartile  $Q_1 = \left(\frac{n+1}{4}\right)$ th term = 3rd term = 12.

(iii) Upper quartile  $Q_3 = \left(\frac{3(n+1)}{4}\right)$ th term = 9th term = 27.

(iv) Interquartile range =  $Q_3 - Q_1 = 27 - 12 = 15$ .

**Answer: (i) 21 (ii) 12 (iii) 27 (iv) 15**

**10 Section B, Question 10 (10 marks)**

- 40.** Find the equation of the line that passes through  $(2, -1)$  and is parallel to the line  $3x + 4y = 7$ . Also write its  $y$ -intercept. **3 marks**

**SOLUTION**

The slope of  $3x + 4y = 7$  is  $-\frac{3}{4}$ , and a parallel line has the same slope.

$y + 1 = -\frac{3}{4}(x - 2)$  gives  $4y + 4 = -3x + 6$ , that is  $3x + 4y - 2 = 0$ .

Put  $x = 0$ :  $y = \frac{1}{2}$ .

**Answer:**  $3x + 4y - 2 = 0$ , with  $y$ -intercept  $\frac{1}{2}$

- 41.** Find  $x$  and  $y$  if  $\begin{bmatrix} 2 & x \\ y & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 8 \\ 7 \end{bmatrix}$ . **3 marks**

**SOLUTION**

Multiplying,  $\begin{bmatrix} 2 + 2x \\ y + 6 \end{bmatrix} = \begin{bmatrix} 8 \\ 7 \end{bmatrix}$ .

So  $2 + 2x = 8$  gives  $x = 3$ , and  $y + 6 = 7$  gives  $y = 1$ .

**Answer:**  $x = 3, y = 1$

- 42.** Using a ruler and compasses only, construct  $\triangle ABC$  with  $AB = 5$  cm,  $BC = 6$  cm and  $\angle ABC = 60^\circ$ . Construct the incircle of the triangle and measure its radius. **4 marks**

**SOLUTION**

Draw  $BC = 6$  cm. At  $B$  construct an angle of  $60^\circ$  and cut off  $BA = 5$  cm on its arm. Join  $AC$ .

Construct the bisectors of  $\angle B$  and  $\angle C$ . They meet at the incentre  $I$ .

From  $I$  draw  $IN$  perpendicular to  $BC$ , meeting it at  $N$ . With centre  $I$  and radius  $IN$  draw the circle. It touches all three sides.

On measuring,  $IN$  comes to about 1.6 cm.

**Answer:** the radius of the incircle measures about 1.6 cm

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